ITEC 50: Web Systems and Technologies

Final Project — Group Progress Update Report

Project Title: DeciMark

Members: Lyra Phasma

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1 Project Overview

DeciMark is a Bookmark Link Manager — a web application that enables users to save, organize, retrieve, and annotate URLs using the **Johnny.Decimal** classification system. The application is server-backed with a full database layer, and requires an account to use. No client-side installation is required — users access DeciMark entirely through a browser, with all complexity contained on the server side.

Target users are students and professionals who need structured, consistent link organization without the entropy of free-form tagging systems. In later development stages, DeciMark is planned to be distributable as a portable installable — `.exe`, `.msi`, or `.AppImage` — for users who prefer self-hosting without a full server setup.

2 Current Progress

Conceptualization and competitive analysis are complete. Core design philosophy and feature differentiation from existing tools — including Pinboard, linkding, and Raindrop.io — have been fully documented. The Johnny.Decimal classification scheme has been selected and justified as the primary organizational model. The project scope has evolved from an initial static concept into a full server-backed implementation, with user accounts and a persistent database layer.

3 Planned Tasks

- Implementation of the card-based UI with list-view toggle
- Johnny.Decimal ID field integration and range-based filtering
- Tag support and multi-attribute search functionality
- User account system and authentication
- Cloud synchronization of bookmark data across sessions
- Persistent database integration
- Final polish, accessibility improvements, and export functionality

- Long-term: portable distribution as `.exe`, `.msi`, and `.AppImage`

4 Tools and Technologies

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Python, FastAPI
- **Database:** PostgreSQL, SQLModel, Alembic (migrations)
- **Authentication:** JWT, itsdangerous, bcrypt
- **Design reference:** Johnny.Decimal organizational system (`https://johnnydecimal.com`)

5 Challenges / Concerns

Working as a sole member required managing all responsibilities — research, design, documentation, and implementation — independently. This was not without its silver linings, however. A concurrent hackathon project provided substantial hands-on experience in full-stack development, database architecture, backend frameworks, and deployment pipelines, all of which have meaningfully deepened my understanding of web systems at a level that directly informs the execution of DeciMark.

6 Additional Notes

It was not all in vain.

The hackathon experience — undertaken in parallel with this course — resulted in the development of **Pebble Recall**, a full-stack psychiatric care companion application built with Python, FastAPI, SQLModel, PostgreSQL, and Jinja2, stored at `https://github.com/phasmolysis-team/phasmolysis-pebble-recall` and deployed live at `https://pebble-recall.lyra-on.top`.

Pebble Recall placed **#1 in judge's choice** and **#1 in community vote** out of more than 140 competing teams, winning the **□20,000 grand prize**.

The skills developed over the course of that project — including database migration with Alembic, asynchronous backend architecture, authentication systems, and full deployment pipelines — are not merely adjacent to DeciMark. They are directly applicable to it. The detour, it turns out, was not a detour at all.

DeciMark is currently in active development, with the backend implementation underway. The project repository is publicly available at `https://github.com/whinee/ITEC50-Finals`. A live deployment will follow as the frontend and backend are integrated.

To promote genuine learning while protecting both intellectual property and academic integrity, the DeciMark repository is multi-licensed:

- **Mozilla Public License 2.0** — applied to files derived from Pebble Recall, in compliance with that project's licensing terms
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